What Happens to the Polarization Vectors: From $\mathcal{M}$ to $\overline{|\mathcal{M}|^2}$, Step by Step

PHYS 7502

1 Where the Polarization Vectors Live in $\mathcal{M}_s$

Start with the s-channel amplitude:

$$\mathcal{M}_s = \frac{-e^2}{s} \underbrace{\varepsilon^*_{3\alpha}}_{\substack{\text{outgoing}\\ \text{photon 3}}} \underbrace{\varepsilon_{2\beta}}_{\substack{\text{incoming}\\ \text{photon 2}}} \bar{u}(4)\,\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\,u(1) \tag{1}$$

Where Each Polarization Comes From

- $\varepsilon_{2\beta}$ (no star) = incoming photon 2. It sits at **Vertex A**, where $\gamma_{\text{in}}(p_2)$ is absorbed. The Lorentz index β contracts with the γ^β at that vertex.
- $\varepsilon^*_{3\alpha}$ (star) = outgoing photon 3. It sits at **Vertex B**, where $\gamma_{\text{out}}(p_3)$ is emitted. The star is there because **outgoing photons always carry** ε^* (this is a Feynman rule). The index α contracts with γ^α.

The polarization vectors are just **4-component complex numbers** ε_μ. They sit *outside* the spinor sandwich as overall multiplicative factors.

2 Where the Polarization Vectors Live in $\mathcal{M}_s^*$

Taking the complex conjugate of $\mathcal{M}_s$:

$$\mathcal{M}_s^* = \frac{-e^2}{s} \underbrace{\varepsilon_{3\rho}}_{\substack{\text{was } \varepsilon^*_{3\alpha}\\ \text{star removed}}} \underbrace{\varepsilon^*_{2\sigma}}_{\substack{\text{was } \varepsilon_{2\beta}\\ \text{star added}}} \bar{u}(1)\,\gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho\,u(4) \tag{2}$$

What Conjugation Does to Polarizations

Every polarization vector gets its star toggled:

- $\varepsilon^*_{3\alpha} \longrightarrow \varepsilon_{3\rho}$ (star removed)
- $\varepsilon_{2\beta} \longrightarrow \varepsilon^*_{2\sigma}$ (star added)

I also relabeled the dummy Lorentz indices: $\alpha \to \rho$ and $\beta \to \sigma$. This is essential — when

we multiply $\mathcal{M}_s \times \mathcal{M}_s^*$, the indices from $\mathcal{M}_s$ and $\mathcal{M}_s^*$ must be **different** so they don't accidentally contract with each other.

3 Forming $|\mathcal{M}_s|^2$: Polarizations Pair Up

Multiplying $\mathcal{M}_s \times \mathcal{M}_s^*$:

$$|\mathcal{M}_s|^2 = \frac{e^4}{s^2} \underbrace{\varepsilon_{3\alpha}^* \, \varepsilon_{3\rho}}_{\substack{\text{photon 3} \\ \text{pair}}} \underbrace{\varepsilon_{2\beta} \, \varepsilon_{2\sigma}^*}_{\substack{\text{photon 2} \\ \text{pair}}}$$

$$\times \left[\bar{u}(4)\,\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\, u(1)\right] \left[\bar{u}(1)\,\gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho\, u(4)\right] \tag{3}$$

Key Observation: Polarizations Form $\varepsilon\varepsilon^*$ Pairs

Look at what happened: each photon's polarization vector from $\mathcal{M}$ paired with its conjugate from $\mathcal{M}^*$:

Photon	**From $\mathcal{M}_s$**	**From $\mathcal{M}_s^*$**	**Product**
Photon 3 (outgoing)	$\varepsilon_{3\alpha}^*$	$\varepsilon_{3\rho}$	$\varepsilon_{3\alpha}^* \, \varepsilon_{3\rho}$
Photon 2 (incoming)	$\varepsilon_{2\beta}$	$\varepsilon_{2\sigma}^*$	$\varepsilon_{2\beta} \, \varepsilon_{2\sigma}^*$

Each pair has one ε and one ε^* with **different** Lorentz indices. This is exactly the structure the polarization sum is designed to handle.

4 Summing Over Polarizations: $\sum \varepsilon_\mu \varepsilon_\nu^* \to -g_{\mu\nu}$

Now we sum over the two physical polarization states of each photon.

4.1 Photon 3 (outgoing)

$$\sum_{\lambda_3=1,2} \varepsilon_{3\alpha}^{(\lambda_3)*} \, \varepsilon_{3\rho}^{(\lambda_3)} \longrightarrow -g_{\alpha\rho} \tag{4}$$

What $-g_{\alpha\rho}$ Does

The metric tensor $-g_{\alpha\rho}$ **contracts the indices α and ρ.** Wherever α and ρ appear in the expression, they become the *same* index (one upper, one lower), and you sum over it:

$$(\cdots)^\alpha(\cdots)_\rho \times (-g_{\alpha\rho}) = -(\cdots)^\alpha(\cdots)_\alpha$$

In our trace, γ^α (from $\mathcal{M}_s$) and γ^ρ (from $\mathcal{M}_s^*$) become γ^α and γ_α — a contracted pair.

4.2 Photon 2 (incoming)

$$\sum_{\lambda_2=1,2} \varepsilon_{2\beta}^{(\lambda_2)}\, \varepsilon_{2\sigma}^{(\lambda_2)*} \;\longrightarrow\; -g_{\beta\sigma} \tag{5}$$

This contracts β with σ: γ^β and γ^σ become γ^β and γ_β.

4.3 The Two Minus Signs

Each polarization sum contributes a factor of (-1). Two photons give $(-1)(-1) = +1$. So there is **no net sign change** from the photon sums.

5 Putting It All Together

Substituting (4) and (5) into (3):

$$\sum_{\text{pol}} |\mathcal{M}_s|^2 = \frac{e^4}{s^2}\,(-g_{\alpha\rho})\,(-g_{\beta\sigma}) \\ \times\; \Big[\bar{u}(4)\,\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\,u(1)\Big]\,\Big[\bar{u}(1)\,\gamma^\sigma(\not{p}_1 + \not{p}_2)\gamma^\rho\,u(4)\Big] \tag{6}$$

The $(-g_{\alpha\rho})$ contracts α with ρ, turning $\gamma^\rho \to \gamma_\alpha$:

$$\gamma^\alpha \cdots \gamma^\rho \;\longrightarrow\; \gamma^\alpha \cdots \gamma_\alpha$$

The $(-g_{\beta\sigma})$ contracts β with σ, turning $\gamma^\sigma \to \gamma_\beta$:

$$\gamma^\beta \cdots \gamma^\sigma \;\longrightarrow\; \gamma^\beta \cdots \gamma_\beta$$

After also applying Casimir's trick (spin sums → trace):

Final Trace: All Polarizations Eliminated

$$\boxed{\frac{1}{4}\sum_{\text{spins,pol}} |\mathcal{M}_s|^2 = \frac{e^4}{4s^2}\,\text{Tr}\Big[\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\,\not{p}_1\,\gamma_\beta(\not{p}_1 + \not{p}_2)\gamma_\alpha\,\not{p}_4\Big]} \tag{7}$$

No polarization vectors remain. The $-g_{\mu\nu}$ sums replaced every $\varepsilon\,\varepsilon^*$ pair with a metric contraction of Lorentz indices. What is left is a pure trace of γ-matrices and momenta.

6 Visual Summary: The Polarization Vector Journey

Complete Flow

Stage 1: In $\mathcal{M}_s$ — polarizations are free 4-vectors with specific indices:

$$\mathcal{M}_s \ni \ \varepsilon^*_{3\alpha}\, \varepsilon_{2\beta}\, \bar{u}(4)\gamma^{\alpha}(\not{p}_1 + \not{p}_2)\gamma^{\beta}u(1)$$

Stage 2: In $\mathcal{M}_s^*$ — stars toggle, indices relabeled:

$$\mathcal{M}_s^* \ni \ \varepsilon_{3\rho}\, \varepsilon^*_{2\sigma}\, \bar{u}(1)\gamma^{\sigma}(\not{p}_1 + \not{p}_2)\gamma^{\rho}u(4)$$

Stage 3: In $|\mathcal{M}_s|^2$ — polarizations pair up:

$$|\mathcal{M}_s|^2 \ni \underbrace{\varepsilon^*_{3\alpha}\varepsilon_{3\rho}}_{\text{photon 3 pair}} \underbrace{\varepsilon_{2\beta}\varepsilon^*_{2\sigma}}_{\text{photon 2 pair}} \times (\text{spinor products})$$

Stage 4: Polarization sum — each pair $\to -g_{\mu\nu}$:

$$\varepsilon^*_{3\alpha}\varepsilon_{3\rho} \to -g_{\alpha\rho}, \qquad \varepsilon_{2\beta}\varepsilon^*_{2\sigma} \to -g_{\beta\sigma}$$

Stage 5: Indices contract — $\gamma^{\rho} \to \gamma_{\alpha}$ and $\gamma^{\sigma} \to \gamma_{\beta}$:

$$\mathrm{Tr}\Big[\gamma^{\alpha}(\not{p}_1 + \not{p}_2)\gamma^{\beta}\, \not{p}_1\, \gamma_{\beta}(\not{p}_1 + \not{p}_2)\gamma_{\alpha}\, \not{p}_4\Big]$$

No ε's remain. The trace has only γ-matrices and momenta — ready for the contraction theorems.

7 Why $-g_{\mu\nu}$ and Not Something Else

The Physics Behind $\sum \varepsilon_\mu \varepsilon^*_\nu \to -g_{\mu\nu}$

A real photon has **2 physical polarization states** (transverse). The exact sum over these two is:

$$\sum_{\lambda=1,2} \varepsilon^{(\lambda)}_{\mu}\varepsilon^{(\lambda)*}_{\nu} = -g_{\mu\nu} + \frac{k_\mu \bar{k}_\nu + \bar{k}_\mu k_\nu}{k \cdot \bar{k}}$$

where $\bar{k}$ is a reference vector. The $-g_{\mu\nu}$ part sums over *all four* polarization states (including unphysical longitudinal and scalar modes). The extra $k_\mu k_\nu$ terms subtract them back out.

But the **Ward identity** says $k^\mu \mathcal{M}_\mu = 0$ — the amplitude vanishes when you replace any external photon polarization with its own momentum. So the $k_\mu k_\nu$ terms give zero when contracted with the amplitude, and we can just use $-g_{\mu\nu}$.

Griffiths' alternative (Eq. 7.105) uses the Coulomb gauge form $\sum \varepsilon_i \varepsilon^*_j = \delta_{ij} - \hat{k}_i \hat{k}_j$, which only has spatial indices ($i, j = 1, 2, 3$). This is correct but frame-dependent and messier. The $-g_{\mu\nu}$ form is Lorentz covariant and much cleaner.